Google Data Analyst Case Study Track A

Cyclistic Bike Converting Casual Riders to Annual Members

Author: Bhagya M

Period Analyzed: October 2025 – September 2026

Total Cleaned Records Analyzed: ~4.1 Million Trips

Executive Summary

Cyclistic's finance analysts have concluded that annual members are significantly more profitable than casual riders. While flexible pricing plans (single-ride passes and full-day passes) attract casual riders, maximizing the number of annual members is critical for long-term sustainable growth.

This report analyzes how annual members and casual riders use Cyclistic bikes differently to design targeted marketing strategies aimed at converting casual riders into annual members.

Primary Goals

Primary Business Goal: Maximize the number of annual memberships to drive long-term revenue growth.

Primary Analyst Goal: Identify how annual members and casual riders use Cyclistic bikes differently across seasonality, weekly behaviors, ride durations, and bike preference.

Key Data Findings

1. Annual & Seasonal Trends

Seasonality Impact: Overall trip volume follows a distinct bell-curve distribution across the year:

Peak Months: Demand peaks in July (565,273 trips) and August (561,330 trips).

Low Months: Demand drops sharply in autumn and hits the lowest point in December (96,158 trips) and January (97,118 trips) due to winter weather.

Member vs. Casual: In overall there are always more annual members than casual riders, annual member - 61.52% and casual members - 38.48% and their different stays consist through the year.

Bike Type Shift: At the beginning of the year, classic bike and e-bike usage was roughly equal. However, leading into the warmer months and second half of the year, both user

groups showed a strong preference shift toward electric bikes (e-bikes). Across the entire 12-month period, e-bikes accounted for 56.6% of all rides.

2. Weekly & Daily Behavioral Differences

Weekday vs. Weekend Usage:

Annual Members: Exhibit strong weekday usage (peaking Tuesday through Thursday), indicating that members predominantly use Cyclistic for daily commuting and functional transportation. Member trip counts decrease slightly on weekends.

Casual Riders: Display the exact opposite trend. Casual trip volume surges significantly on weekends (Saturday and Sunday), highlighting clear recreational and leisure usage.

Ride Duration:

Casual riders consistently log significantly longer average trip durations than annual members across all days of the week (averaging ~19–23 minutes on weekends vs. ~13–14 minutes for members).

Trip durations for both groups increase on weekends, but casual riders show the highest jump in ride length.

Bike Type vs. Duration:

While e-bikes are used more frequently overall, classic bikes consistently register higher average trip durations across both user types, suggesting classic bikes are preferred for longer leisure rides while e-bikes are favored for speed and efficiency.

Summary Comparison Matrix

Metric / Dimension	Annual Members	Casual Riders
Primary Use Case	Work commuting & functional transport	Leisure, tourism, & weekend recreation
Peak Days	Mid-week (Tuesday – Thursday)	Weekends (Saturday & Sunday)
Trip Duration	Shorter, consistent (~11–13 mins)	Longer, variable (~18–23 mins)
Seasonality Risk	Drops in value	Drops in value
Bike Preference	E-bikes (~58%) > Classic bikes	E-bikes (~59%) > Classic bikes

High-Level Recommendations (Act Phase)

To effectively convert casual riders into annual members, marketing strategies should target leisure-oriented weekend riders and offer incentives tailored to their usage habits:

Weekend & Summer Targeted Marketing Campaigns:

Launch digital and on-station marketing campaigns during peak recreational times (Fridays through Sundays from May to August).

Focus messaging on the long-term cost savings of an annual membership for regular weekend riders.

Seasonal / Seasonal-to-Annual Promotion ("Flex-to-Annual"):

Introduce a seasonal promotion or credit system where casual pass purchases made during summer months can be credited toward the purchase of a full annual membership.

E-Bike Membership Perks:

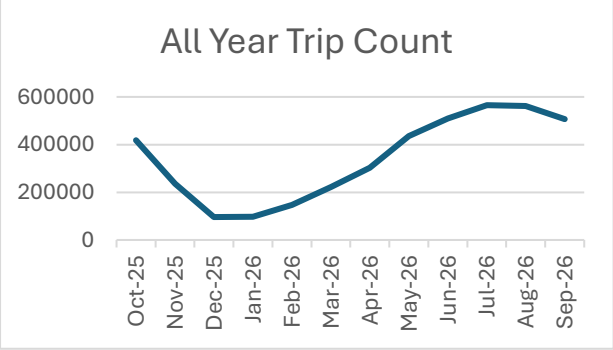
Since e-bikes are preferred by both groups, offer member-exclusive perks for e-bikes, such as discounted per-minute rates or priority reservation windows during peak summer weekends.

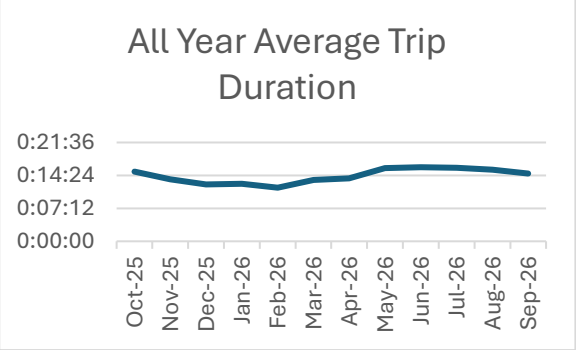
In-App Milestones for Long Duration Riders:

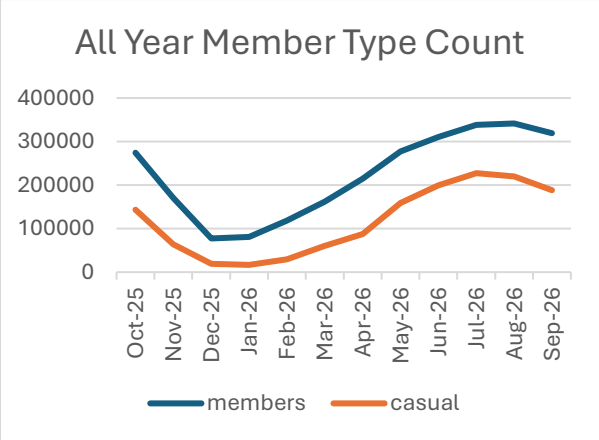
Trigger targeted push notifications and membership conversion offers in the Cyclistic app when a casual rider completes multiple rides over 20 minutes in a single month.

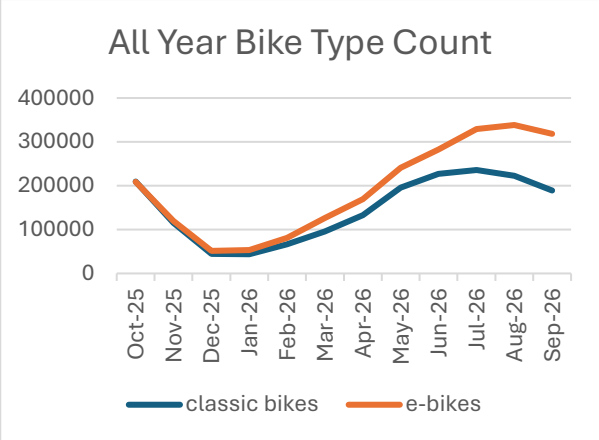
Analyst Supporting Graphs

Annually:



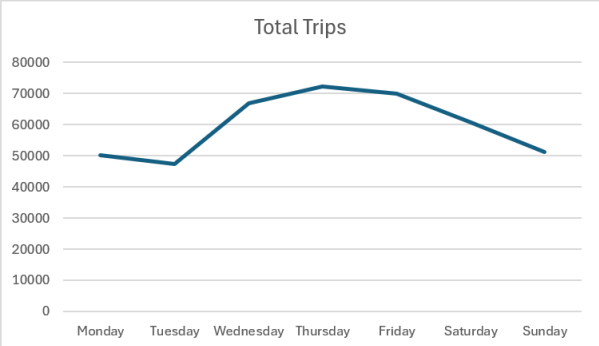


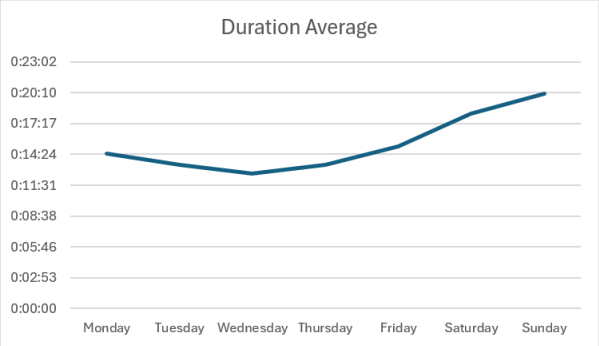


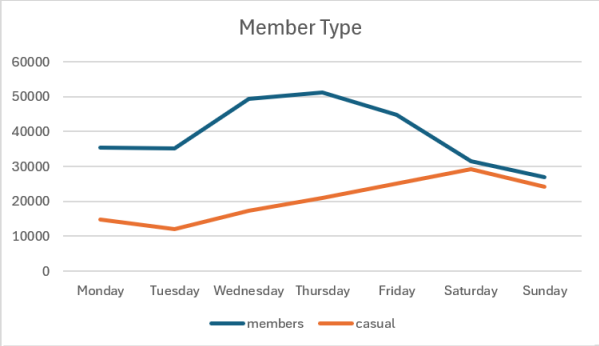


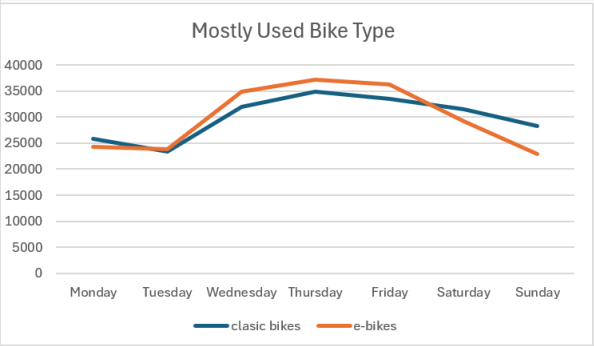
Monthly:

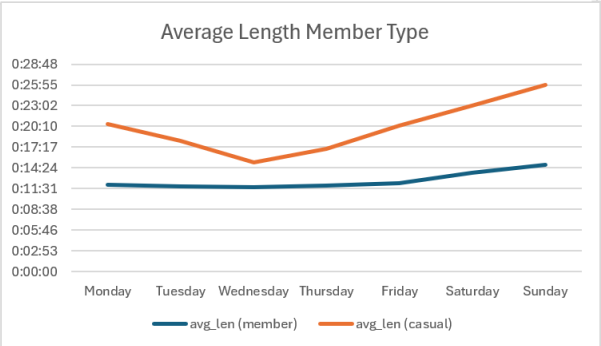
Oct – 2025

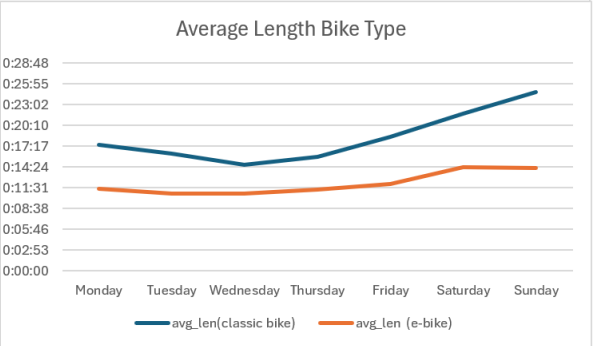




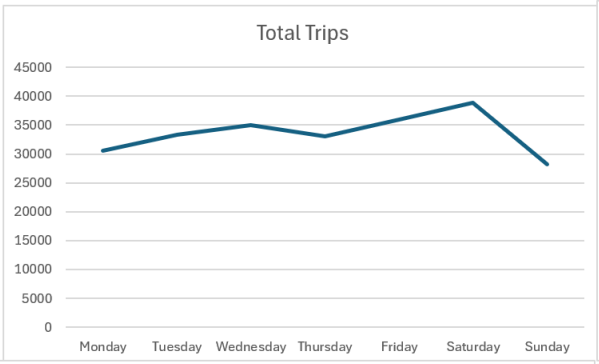


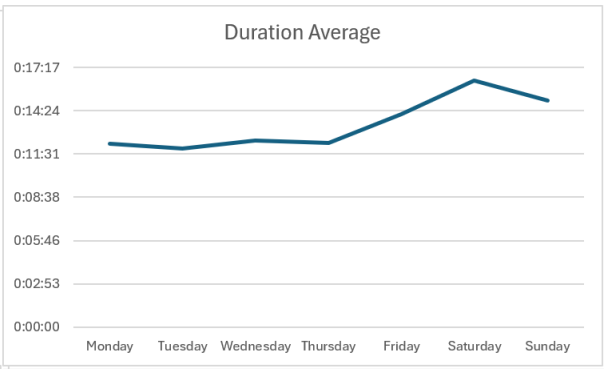


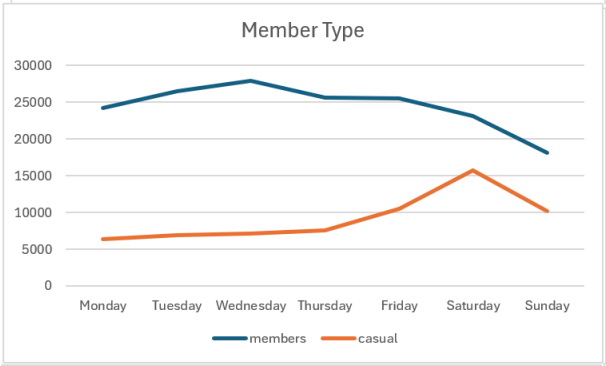


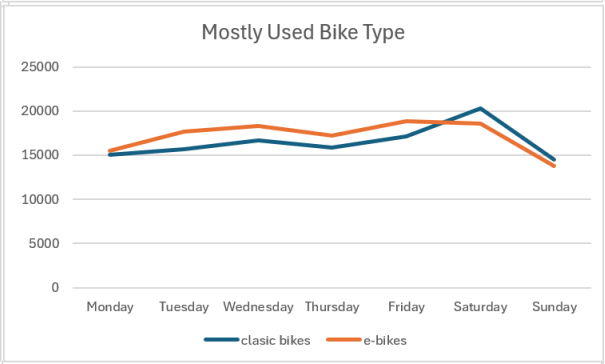


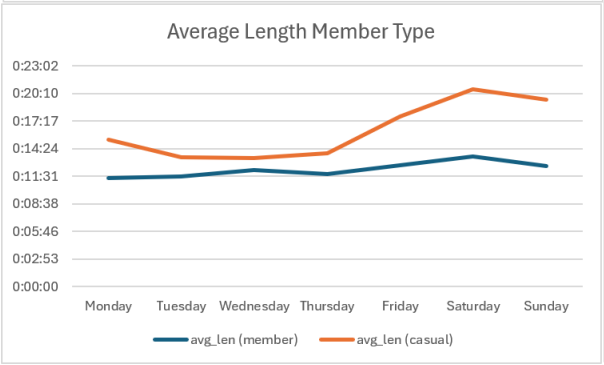
Nov – 2025n

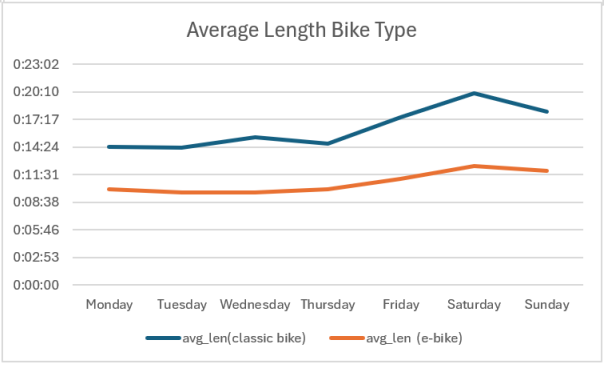

Total Trips
45000
40000
35000
30000
25000
20000
15000
10000
5000
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday


Duration Average
0:17:17
0:14:24
0:11:31
0:08:38
0:05:46
0:02:53
0:00:00
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday

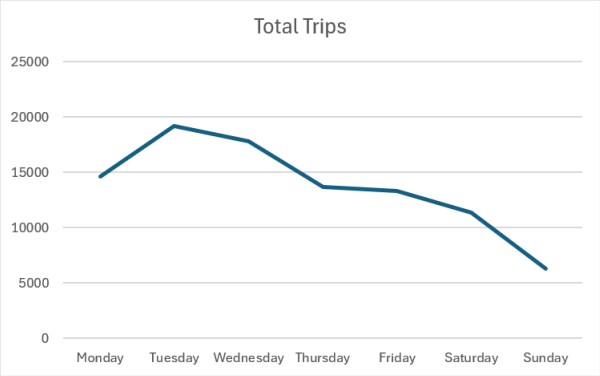

Member Type
30000
25000
20000
15000
10000
5000
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
members
casual

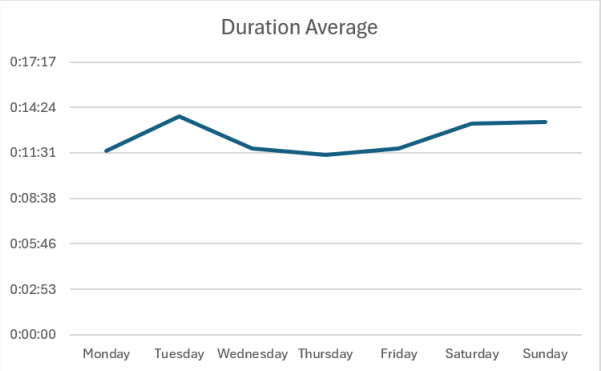

Mostly Used Bike Type
25000
20000
15000
10000
5000
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
clasic bikes
e-bikes

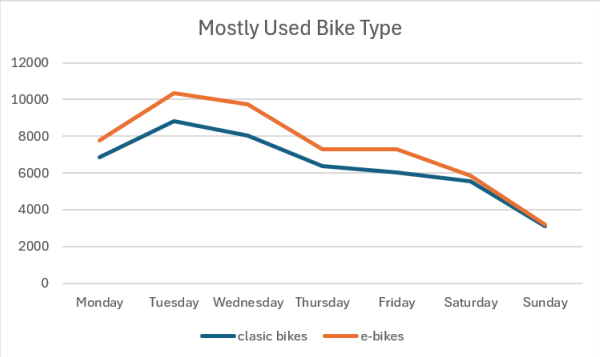

Average Length Member Type
0:23:02
0:20:10
0:17:17
0:14:24
0:11:31
0:08:38
0:05:46
0:02:53
0:00:00
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
avg_len (member)
avg_len (casual)

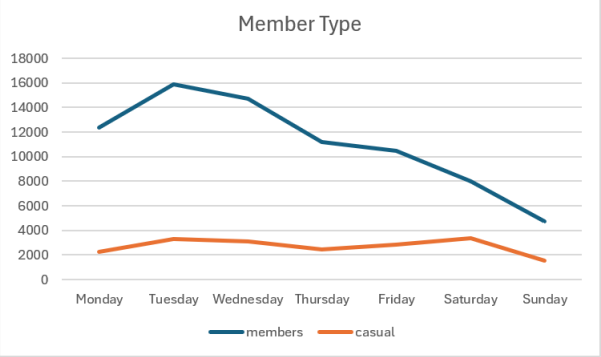

Average Length Bike Type
0:23:02
0:20:10
0:17:17
0:14:24
0:11:31
0:08:38
0:05:46
0:02:53
0:00:00
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
avg_len(classic bike)
avg_len (e-bike)

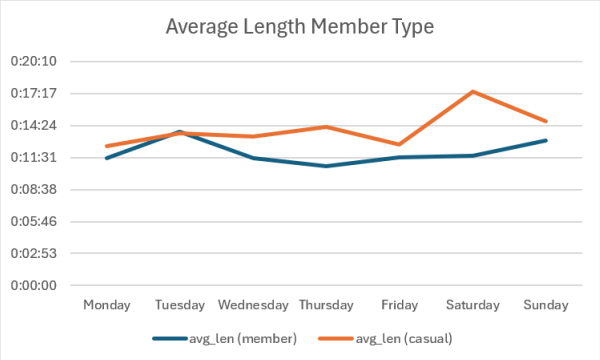
Dec -2025

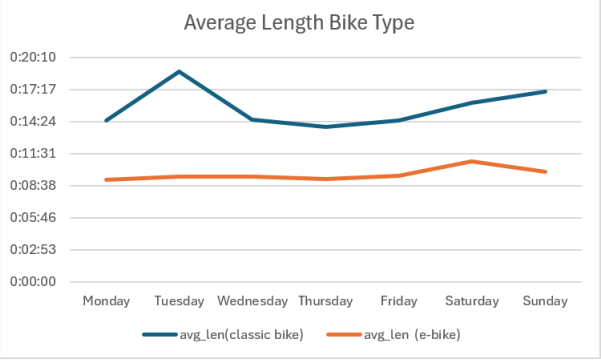

Total Trips
25000
20000
15000
10000
5000
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday


Duration Average
0:17:17
0:14:24
0:11:31
0:08:38
0:05:46
0:02:53
0:00:00
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday

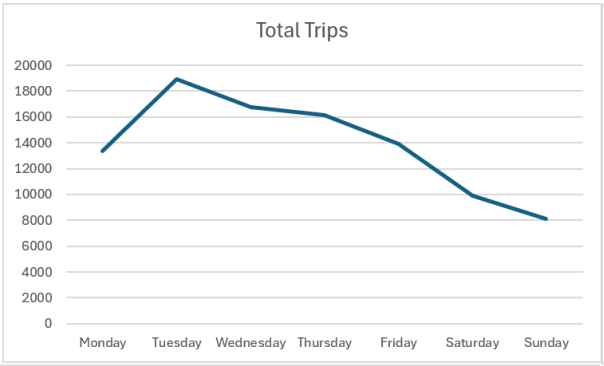

Mostly Used Bike Type
12000
10000
8000
6000
4000
2000
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
clasic bikes
e-bikes

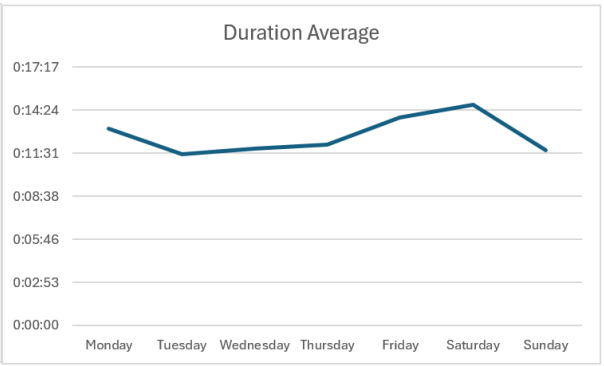

Member Type
18000
16000
14000
12000
10000
8000
6000
4000
2000
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
members
casual

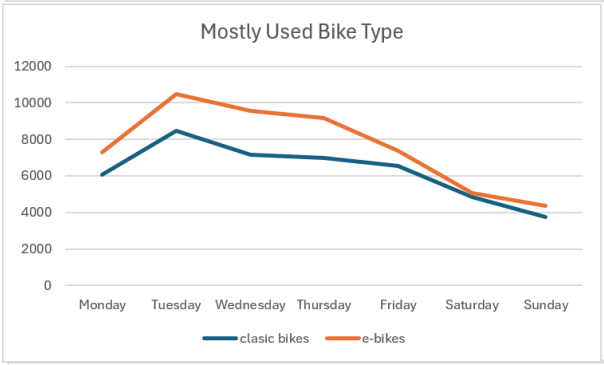

Average Length Member Type
0:20:10
0:17:17
0:14:24
0:11:31
0:08:38
0:05:46
0:02:53
0:00:00
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
avg_len (member)
avg_len (casual)

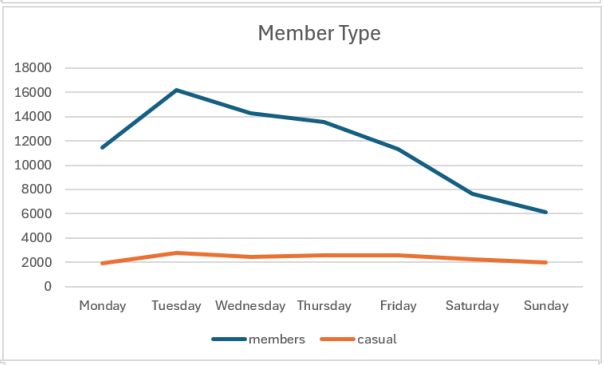

Average Length Bike Type
0:20:10
0:17:17
0:14:24
0:11:31
0:08:38
0:05:46
0:02:53
0:00:00
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
avg_len(classic bike)
avg_len (e-bike)

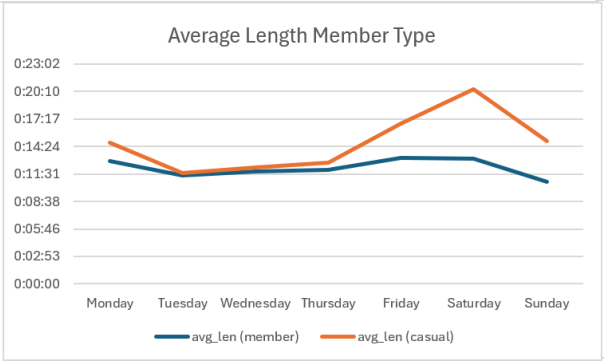
Jan – 2026

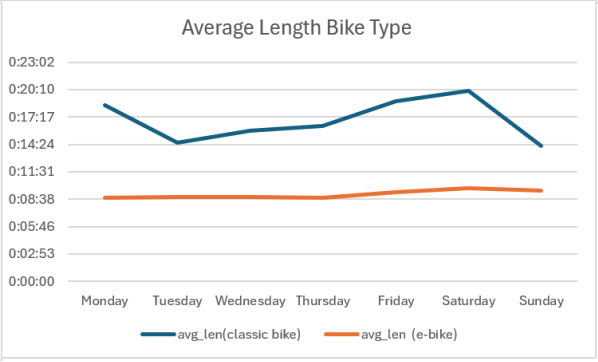

Total Trips
20000
18000
16000
14000
12000
10000
8000
6000
4000
2000
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday


Duration Average
0:17:17
0:14:24
0:11:31
0:08:38
0:05:46
0:02:53
0:00:00
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday

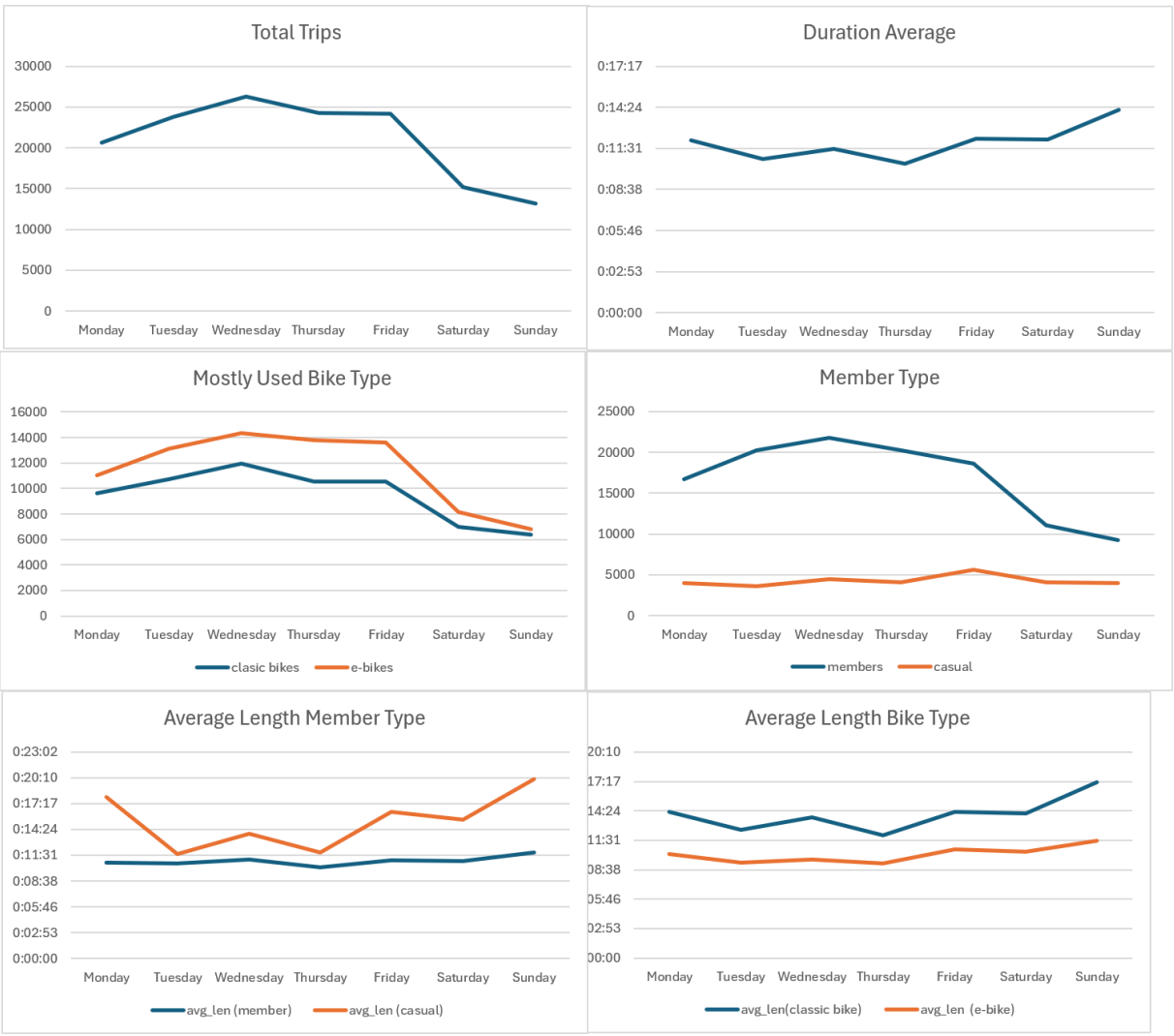

Mostly Used Bike Type
12000
10000
8000
6000
4000
2000
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
clasic bikes
e-bikes


Member Type
18000
16000
14000
12000
10000
8000
6000
4000
2000
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
members
casual

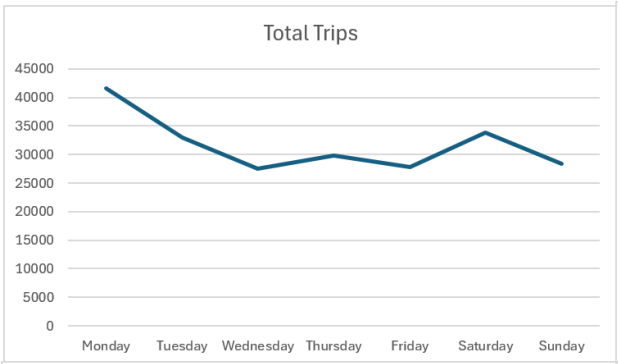

Average Length Member Type
0:23:02
0:20:10
0:17:17
0:14:24
0:11:31
0:08:38
0:05:46
0:02:53
0:00:00
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
avg_len (member)
avg_len (casual)

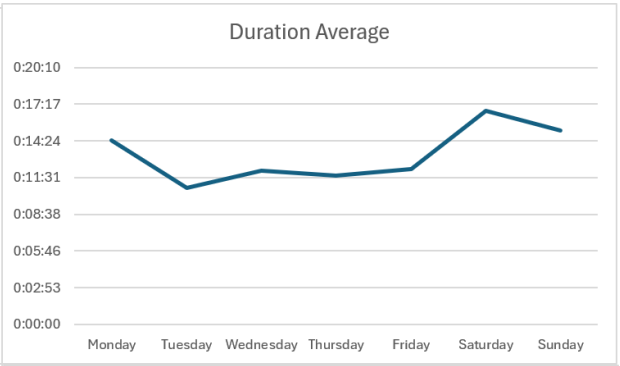

Average Length Bike Type
0:23:02
0:20:10
0:17:17
0:14:24
0:11:31
0:08:38
0:05:46
0:02:53
0:00:00
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
avg_len(classic bike)
avg_len (e-bike)

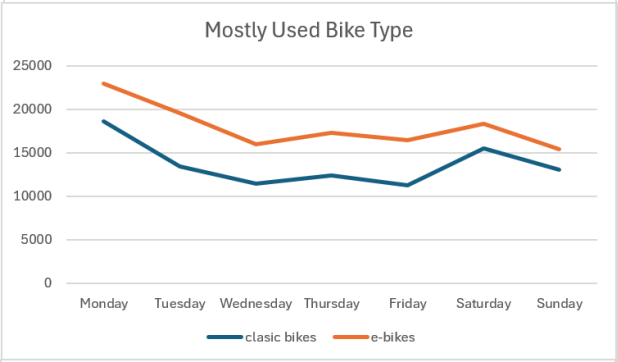
Feb – 2026

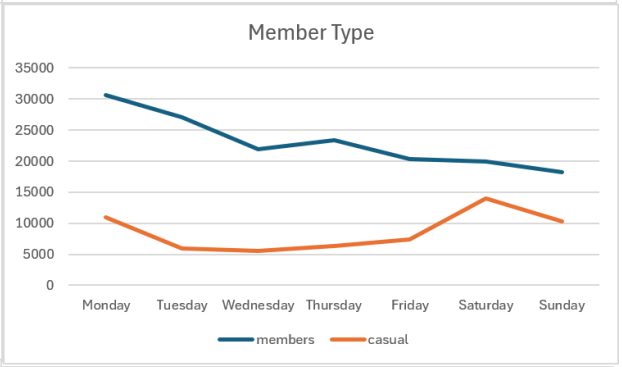

Total Trips
30000
25000
20000
15000
10000
5000
0
Monday Tuesday Wednesday Thursday Friday Saturday Sunday
Duration Average
0:17:17
0:14:24
0:11:31
0:08:38
0:05:46
0:02:53
0:00:00
Monday Tuesday Wednesday Thursday Friday Saturday Sunday
Mostly Used Bike Type
16000
14000
12000
10000
8000
6000
4000
2000
0
Monday Tuesday Wednesday Thursday Friday Saturday Sunday
clasic bikes
e-bikes
Member Type
25000
20000
15000
10000
5000
0
Monday Tuesday Wednesday Thursday Friday Saturday Sunday
members
casual
Average Length Member Type
0:23:02
0:20:10
0:17:17
0:14:24
0:11:31
0:08:38
0:05:46
0:02:53
0:00:00
Monday Tuesday Wednesday Thursday Friday Saturday Sunday
avg_len (member)
avg_len (casual)
Average Length Bike Type
20:10
17:17
14:24
11:31
08:38
05:46
02:53
00:00
Monday Tuesday Wednesday Thursday Friday Saturday Sunday
avg_len(classic bike)
avg_len (e-bike)

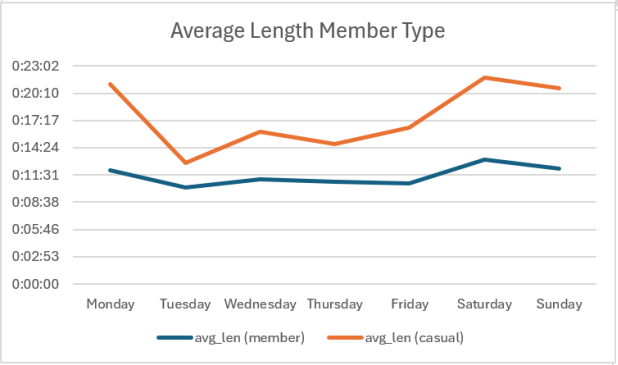
Mar – 2026

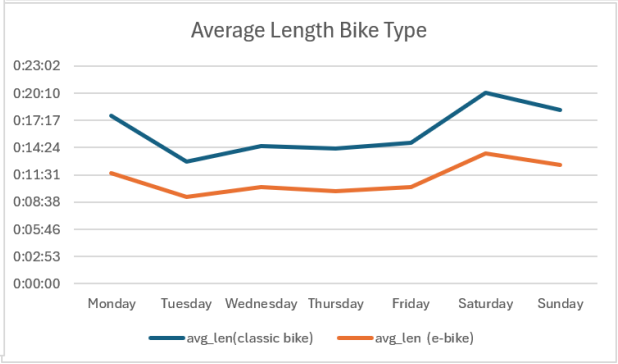

Total Trips
45000
40000
35000
30000
25000
20000
15000
10000
5000
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday


Duration Average
0:20:10
0:17:17
0:14:24
0:11:31
0:08:38
0:05:46
0:02:53
0:00:00
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday

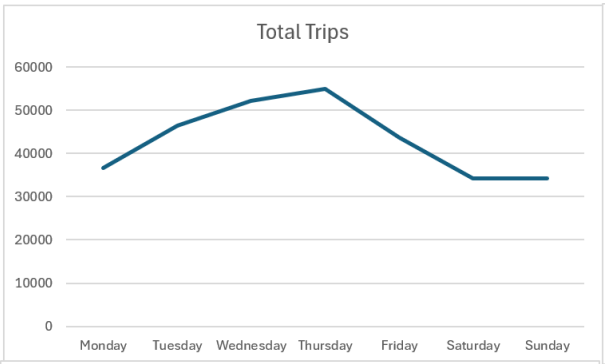

Mostly Used Bike Type
25000
20000
15000
10000
5000
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
clasic bikes
e-bikes

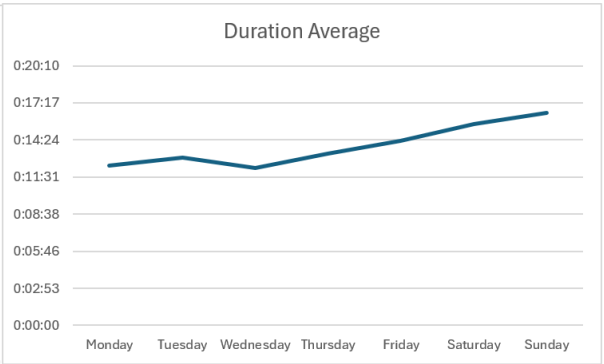

Member Type
35000
30000
25000
20000
15000
10000
5000
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
members
casual

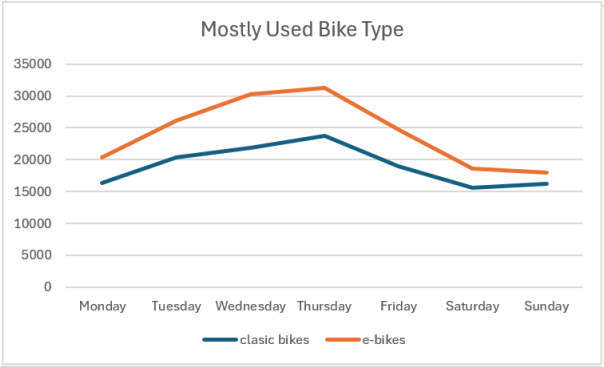

Average Length Member Type
0:23:02
0:20:10
0:17:17
0:14:24
0:11:31
0:08:38
0:05:46
0:02:53
0:00:00
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
avg_len (member)
avg_len (casual)

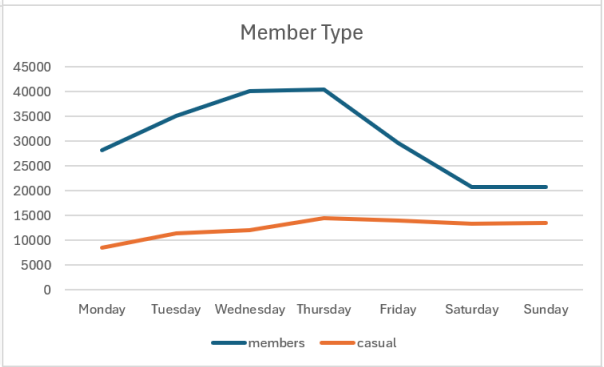

Average Length Bike Type
0:23:02
0:20:10
0:17:17
0:14:24
0:11:31
0:08:38
0:05:46
0:02:53
0:00:00
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
avg_len(classic bike)
avg_len (e-bike)

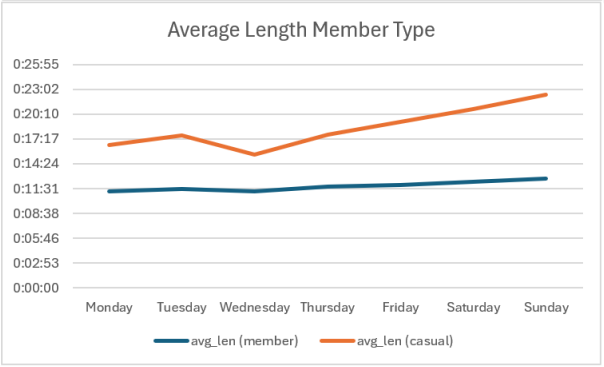
April – 2026

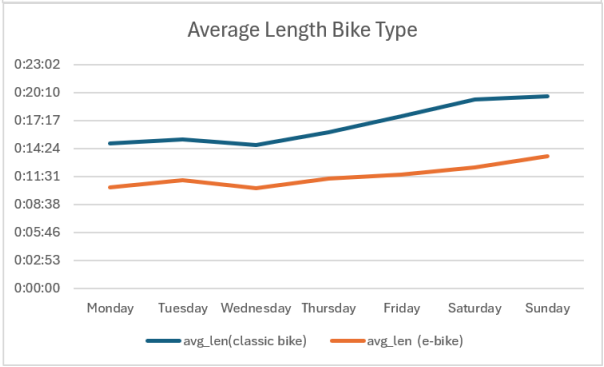

Total Trips
60000
50000
40000
30000
20000
10000
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday


Duration Average
0:20:10
0:17:17
0:14:24
0:11:31
0:08:38
0:05:46
0:02:53
0:00:00
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday

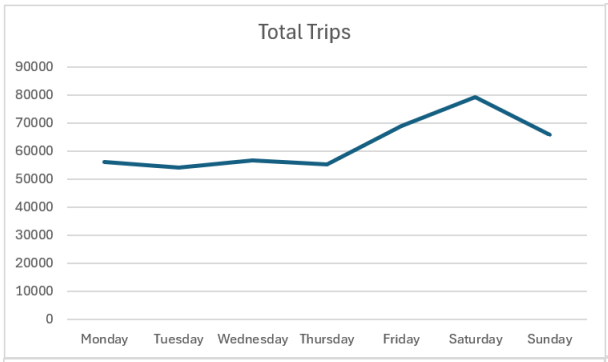

Mostly Used Bike Type
35000
30000
25000
20000
15000
10000
5000
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
clasic bikes
e-bikes

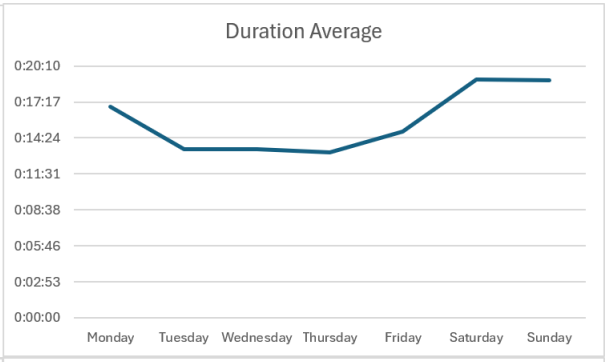

Member Type
45000
40000
35000
30000
25000
20000
15000
10000
5000
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
members
casual

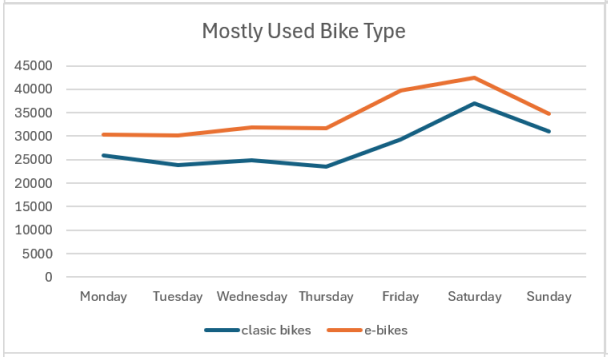

Average Length Member Type
0:25:55
0:23:02
0:20:10
0:17:17
0:14:24
0:11:31
0:08:38
0:05:46
0:02:53
0:00:00
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
avg_len (member)
avg_len (casual)

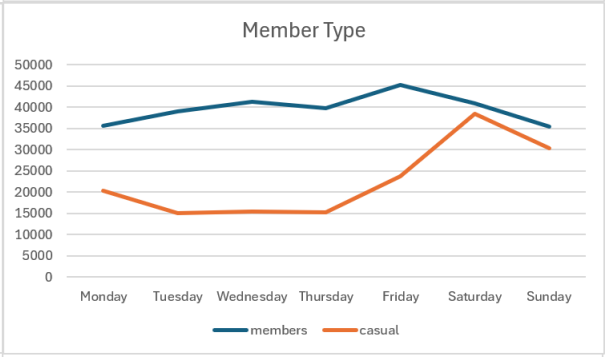

Average Length Bike Type
0:23:02
0:20:10
0:17:17
0:14:24
0:11:31
0:08:38
0:05:46
0:02:53
0:00:00
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
avg_len(classic bike)
avg_len (e-bike)

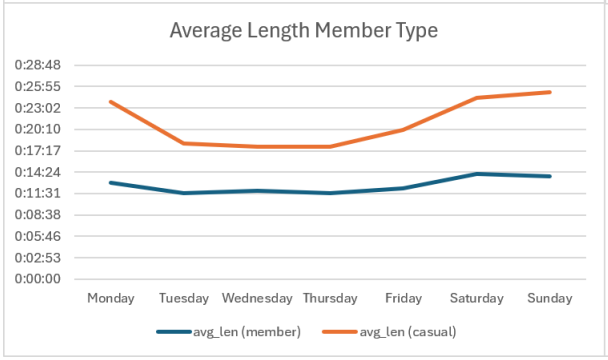
May – 2026

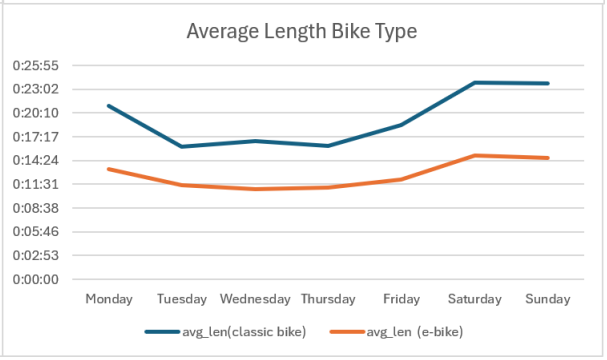

Total Trips
90000
80000
70000
60000
50000
40000
30000
20000
10000
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday


Duration Average
0:20:10
0:17:17
0:14:24
0:11:31
0:08:38
0:05:46
0:02:53
0:00:00
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday

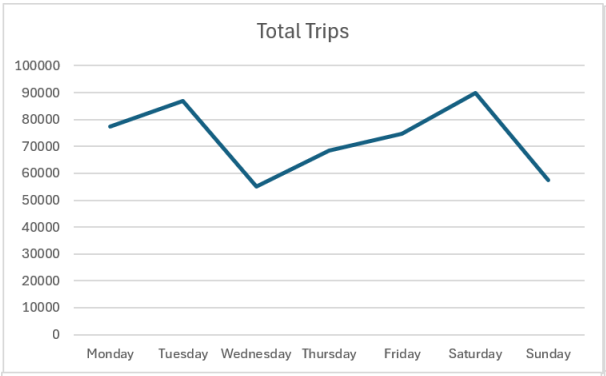

Mostly Used Bike Type
45000
40000
35000
30000
25000
20000
15000
10000
5000
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
clasic bikes
e-bikes

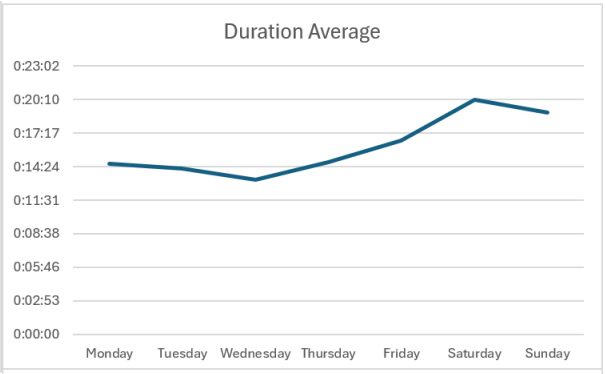

Member Type
50000
45000
40000
35000
30000
25000
20000
15000
10000
5000
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
members
casual

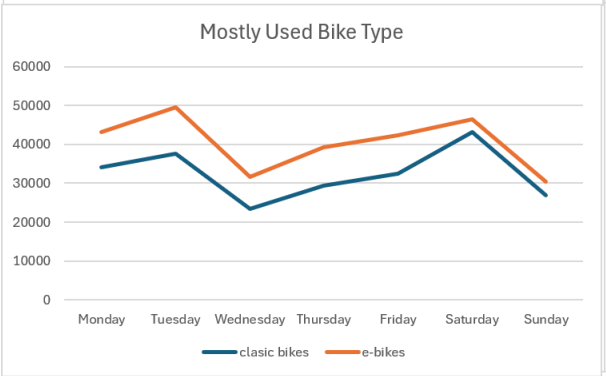

Average Length Member Type
0:28:48
0:25:55
0:23:02
0:20:10
0:17:17
0:14:24
0:11:31
0:08:38
0:05:46
0:02:53
0:00:00
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
avg_len (member)
avg_len (casual)

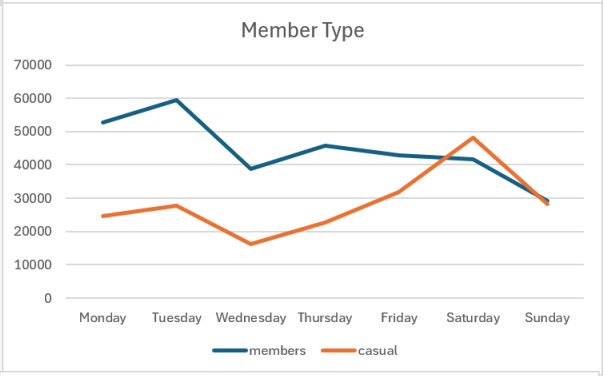

Average Length Bike Type
0:25:55
0:23:02
0:20:10
0:17:17
0:14:24
0:11:31
0:08:38
0:05:46
0:02:53
0:00:00
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
avg_len(classic bike)
avg_len (e-bike)

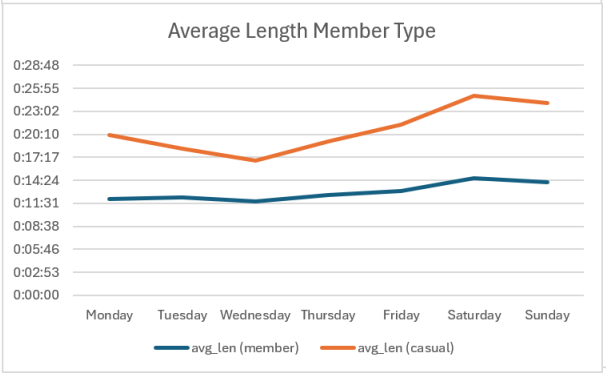
Jun – 2026

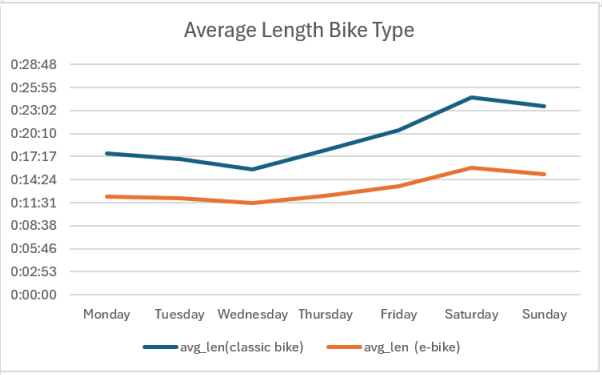

Total Trips
100000
90000
80000
70000
60000
50000
40000
30000
20000
10000
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday


Duration Average
0:23:02
0:20:10
0:17:17
0:14:24
0:11:31
0:08:38
0:05:46
0:02:53
0:00:00
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday

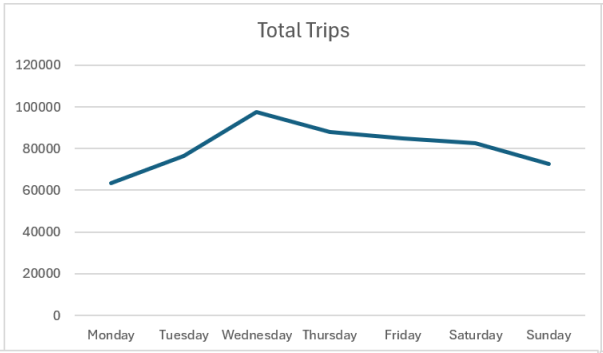

Mostly Used Bike Type
60000
50000
40000
30000
20000
10000
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
clasic bikes
e-bikes

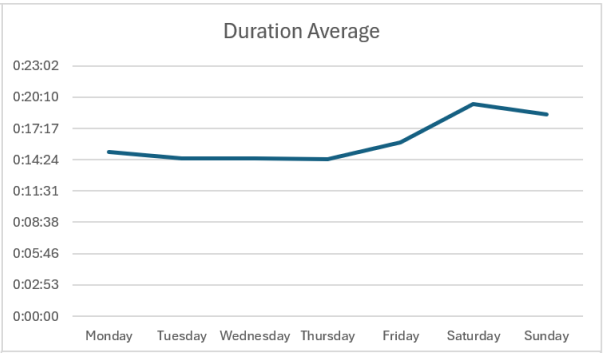

Member Type
70000
60000
50000
40000
30000
20000
10000
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
members
casual

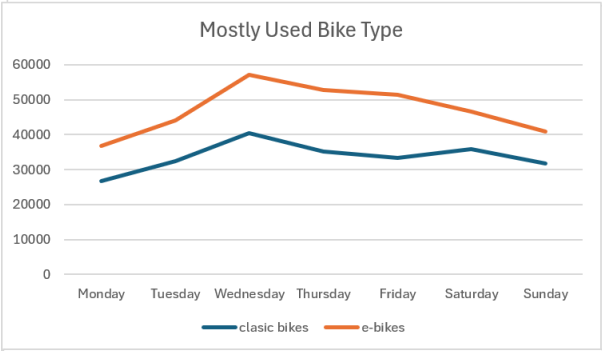

Average Length Member Type
0:28:48
0:25:55
0:23:02
0:20:10
0:17:17
0:14:24
0:11:31
0:08:38
0:05:46
0:02:53
0:00:00
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
avg_len (member)
avg_len (casual)

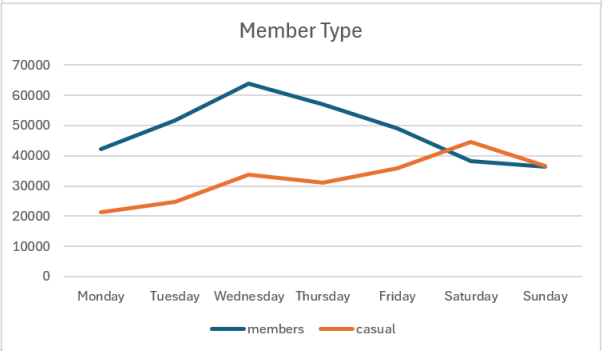

Average Length Bike Type
0:28:48
0:25:55
0:23:02
0:20:10
0:17:17
0:14:24
0:11:31
0:08:38
0:05:46
0:02:53
0:00:00
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
avg_len(classic bike)
avg_len (e-bike)

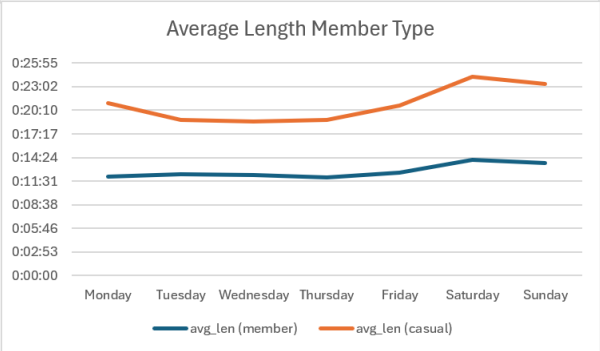
Jul – 2026

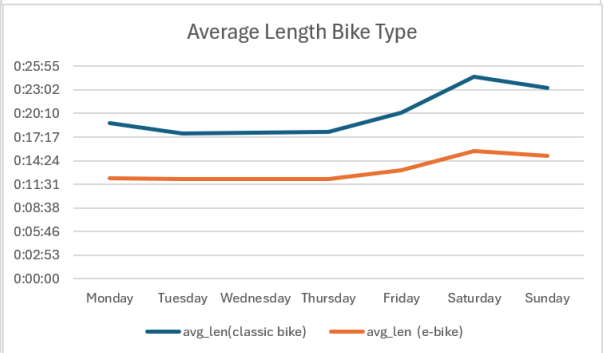

Total Trips
120000
100000
80000
60000
40000
20000
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday


Duration Average
0:23:02
0:20:10
0:17:17
0:14:24
0:11:31
0:08:38
0:05:46
0:02:53
0:00:00
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday

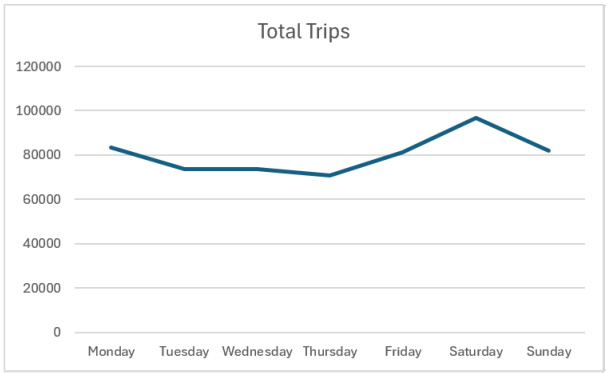

Mostly Used Bike Type
60000
50000
40000
30000
20000
10000
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
clasic bikes
e-bikes

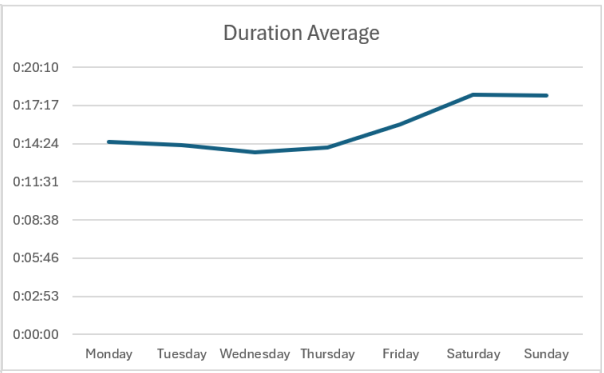

Member Type
70000
60000
50000
40000
30000
20000
10000
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
members
casual

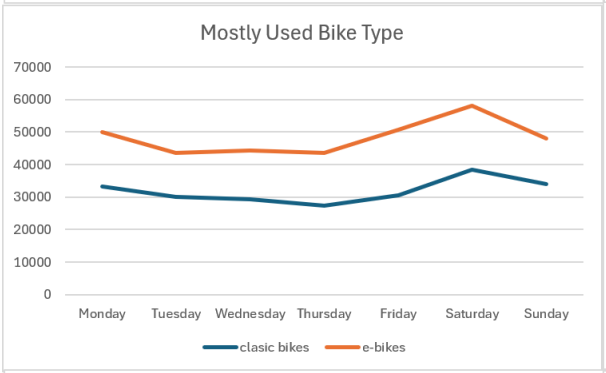

Average Length Member Type
0:25:55
0:23:02
0:20:10
0:17:17
0:14:24
0:11:31
0:08:38
0:05:46
0:02:53
0:00:00
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
avg_len (member)
avg_len (casual)

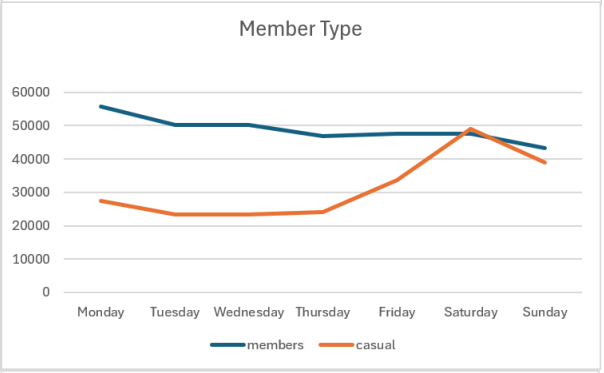

Average Length Bike Type
0:25:55
0:23:02
0:20:10
0:17:17
0:14:24
0:11:31
0:08:38
0:05:46
0:02:53
0:00:00
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
avg_len(classic bike)
avg_len (e-bike)

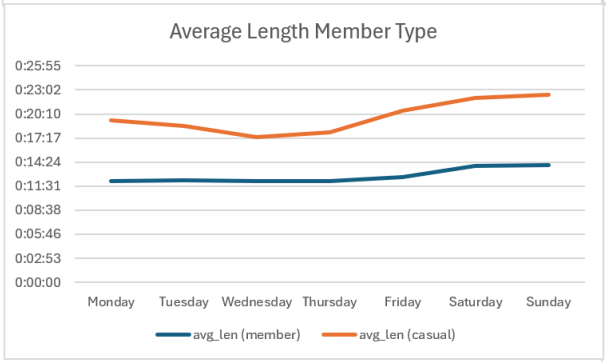
Aug – 2026

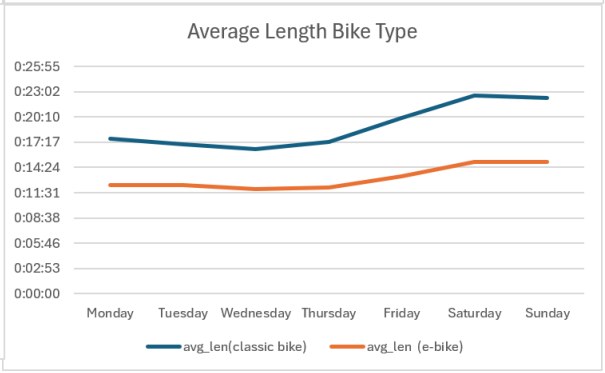

Total Trips
120000
100000
80000
60000
40000
20000
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday


Duration Average
0:20:10
0:17:17
0:14:24
0:11:31
0:08:38
0:05:46
0:02:53
0:00:00
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday

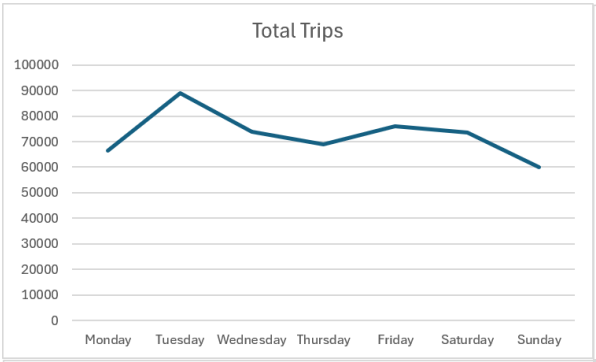

Mostly Used Bike Type
70000
60000
50000
40000
30000
20000
10000
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
clasic bikes
e-bikes

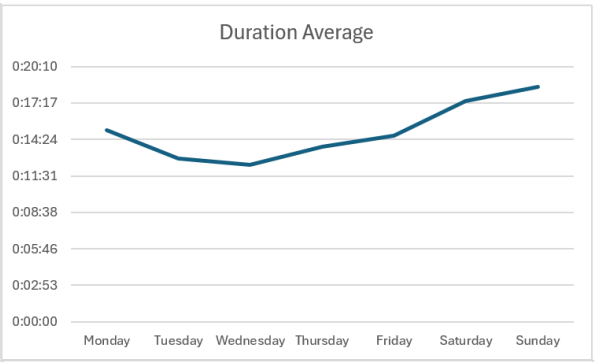

Member Type
60000
50000
40000
30000
20000
10000
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
members
casual

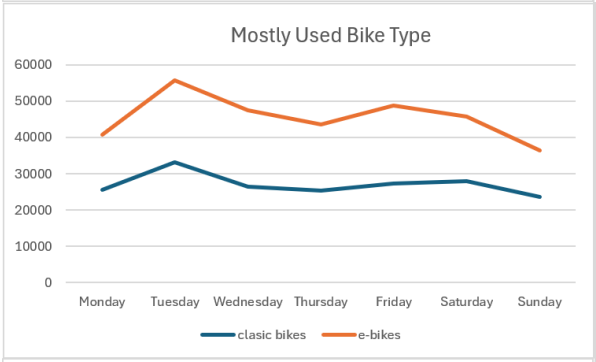

Average Length Member Type
0:25:55
0:23:02
0:20:10
0:17:17
0:14:24
0:11:31
0:08:38
0:05:46
0:02:53
0:00:00
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
avg_len (member)
avg_len (casual)

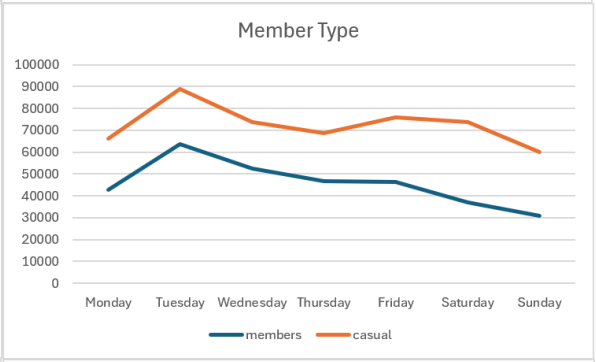

Average Length Bike Type
0:25:55
0:23:02
0:20:10
0:17:17
0:14:24
0:11:31
0:08:38
0:05:46
0:02:53
0:00:00
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
avg_len(classic bike)
avg_len (e-bike)

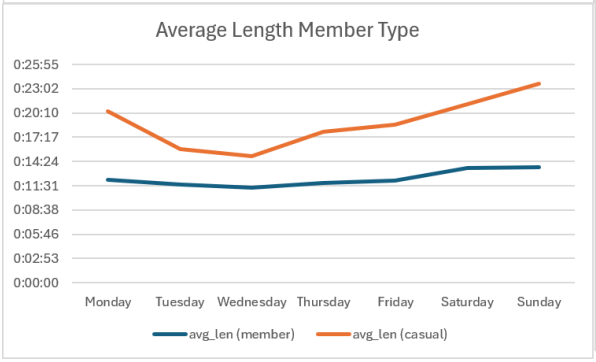
Sep – 2026

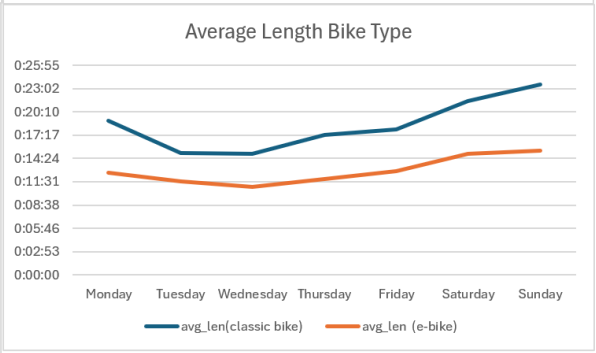

Total Trips
100000
90000
80000
70000
60000
50000
40000
30000
20000
10000
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday


Duration Average
0:20:10
0:17:17
0:14:24
0:11:31
0:08:38
0:05:46
0:02:53
0:00:00
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday


Mostly Used Bike Type
60000
50000
40000
30000
20000
10000
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
clasic bikes
e-bikes


Member Type
100000
90000
80000
70000
60000
50000
40000
30000
20000
10000
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
members
casual


Average Length Member Type
0:25:55
0:23:02
0:20:10
0:17:17
0:14:24
0:11:31
0:08:38
0:05:46
0:02:53
0:00:00
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
avg_len (member)
avg_len (casual)


Average Length Bike Type
0:25:55
0:23:02
0:20:10
0:17:17
0:14:24
0:11:31
0:08:38
0:05:46
0:02:53
0:00:00
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
avg_len(classic bike)
avg_len (e-bike)